Sahaj Bhadja

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1 Summary

Systems software engineer pursuing an MSCS at USC with experience building **Python** test automation frameworks, validation infrastructure, and hardware/software integration tooling. Strong in **Git**, debugging, observability, and platform-based automation for firmware-style system validation.

2 Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Computer Networks, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Computer Networks

3 Experience

USC Dornsife **Los Angeles, CA**
Systems Software Engineer - Test Infrastructure *September 2025 – Present*
- Built validation frameworks and automated reconciliation checks for production pipelines serving **500+ users**, improving observability and issue detection across **5K+ monthly jobs** with **99%+ release reliability**.
- Translated validation requirements into maintainable **Python** tooling with documented APIs and CI/CD workflows, supporting root-cause analysis during architecture discussions and cross-team code reviews.

Grad8 **Los Angeles, CA**
Test Automation Engineer Intern *June 2025 – August 2025*
- Automated release qualification for **Python** services handling **10K+ daily requests**, building debug tooling and GitHub Actions pipelines that reduced manual turnaround by **60%** with documented test workflows.

Aasma Technology Solutions **India**
Systems Software Engineer - Test & Automation - (SecureSync) *August 2023 – July 2024*
- Developed automated test frameworks with **20+ PyTest suites** and nightly Jenkins pipelines across **Linux**, **Windows**, and **macOS** VMs, increasing automation coverage and release qualification confidence by **68%**.
- Created platform automation for issue replication, CSV reporting, and debug workflows, migrating script-based validation toward robust infrastructure while partnering with engineers on NPI-style test coverage in code reviews.

Aasma Technology Solutions **India**
Test Automation Engineer Intern - (VibrantChem) *May 2023 – August 2023*
- Implemented **Jest**, **Mocha**, and **Selenium** automation across modules syncing **2M+ records**, reducing manual effort through Docker/Jenkins workflows with structured logging and documented test APIs.

MedDevice Test Systems **India**
Systems Software Engineer Intern *January 2023 – April 2023*
- Supported **embedded system validation** with simulators and **Python/C++** test libraries, assisting root-cause analysis across hardware/software workflows and maintaining test hardware/software setups for release qualification.

4 Skills

Automation: Python, test frameworks, test infrastructure, firmware validation (familiar), system-level validation, PyTest, Jenkins, CI/CD, release qualification, observability, logging, debugging, root-cause analysis, documentation, Git, code reviews
Embedded & HW: C, C++, C# (familiar), hardware/software integration, UART (familiar), I2C (familiar), SPI (familiar), register interfaces (familiar), oscilloscopes (familiar), multimeters (familiar), signal processing (familiar), optics (familiar), Linux, Windows, macOS, Agile

5 Projects

ControlsSim | *Python, C++, Embedded Validation, Simulators*
- Built a control-loop simulator with **Python/C++** automated scripts validating state transitions, fault handling, and register-style configuration for embedded system validation.

CPU-scheduling Simulator Live | *C++, System-Level Tooling*
- Implemented **C++** scheduling algorithms with automated correctness tests across **1K+ combinations**, reducing debugging time by **20%** through repeatable system-level validation.

OptoBench | *Python, Hardware Interface, Instrument Control*
- Developed **Python** tooling to automate instrument read/write sequences and protocol-based device checks with structured debug logging for hardware/software interface testing.